CCC-C-438c
February 12, 1968
SUPERSEDING
Fed. Spec. CCC-C-438b
August 19, 1965

FEDERAL SPECIFICATION

CLOTH, BUCKRAM, COTTON

This specification was approved by the commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for cotton buckram.

1.2 Classification.

1.2.1 Types and classes. The finished cloth shall be of the following types and classes, as specified (see 6.2).

Type I - Resin finish.
  Class 1 - 5.7 ounces per square yard (fine texture).
  Class 2 - 5.7 ounces per square yard (medium texture).
Type II - Non-chlorine retentive resin finish.
  Class 1 - 4.5 ounces per square yard.
Type III - Starch finish.
  Class 1 - 3.8 ounces per square yard.
  Class 2 - 12.0 ounces per square yard.
Type IV - Starch finish, double texture (2 plies of type III, class 2).
  Class 1 - 24.0 ounces per square yard.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

CCC-T-191 - Textile Test Methods.
PPP-P-1134 - Packaging and Packing of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics).

Federal Standard

Fed. Std. No. 4 - Glossary of Fabric Imperfections.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Rules and Regulations Under the Textile Fiber Products Identification Act.

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D. C. 20580.)

3. REQUIREMENTS

3.1 Standard sample. When a standard sample is available, the finished cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Material.

3.2.1 Yarn. The yarn shall be made from cotton which has been cleaned, carded, drawn and spun into singles yarn for both warp and filling.

3.3 Color. The color shall be natural, bleached or dyed as specified, (see 6.2). When natural is specified, the cloth shall be fabricated from unbleached yarn. The use of reused or dyed fiber will be allowed, subject to the colorfastness tests specified in 3.3.2. The use of sulfur dyes or dyes containing elementary sulfur compounds capable of oxidation to sulfuric acid is prohibited. The dyestuffs shall be chosen and applied so that the dyed cloth shall show no more labile sulfur than shown by the standard sample when tested as specified in 4.4. When no standard sample is available there shall be a minimum amount of staining of the test paper when tested as specified in 4.4.

3.3.1 Matching. When a standard sample is available, the dyed cloth shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.3.2 Colorfastness.

3.3.2.1 Natural cloth. When dyed fibers are utilized (see 3.3), the finished cloth shall show colorfastness to laundering, perspiration, and wet drycleaning equal to or better than the standard sample (see 6.3). When the standard sample is not established for colorfastness, the natural and finished cloth (when composed of dyed fibers) shall show "good" colorfastness to laundering, perspiration, and wet drycleaning when tested as specified in 4.4.

3.3.2.2 Dyed cloth. The dyed and finished cloth shall show colorfastness to laundering, crocking, perspiration, and wet drycleaning equal to or better than the standard sample (see 6.3). When the standard sample is not established for colorfastness, the dyed and finished cloth shall show "good" colorfastness to laundering, perspiration, and wet drycleaning and shall show a Munsell Value for crocking not lower than 8.5. Tests shall be as specified in 4.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.

3.4 Physical requirements. The finished cloth shall conform to the requirements shown in table I when tested as specified in 4.4.

TABLE I. Physical requirements

Type	Class	Weight per sq. yd. ozs. minimum	Yarns per inch minimum		Breaking strength (pounds) minimum		Stiffness (inch pounds) minimum		Shrinkage (percent) maximum		Non-fibrous materials maximum
			Warp	Filling	Warp	Filling	Warp	Filling	Warp	Filling	Percent
I	1	5.7	52	44	63	54	0.020	0.010	3.0	2.0	25
	2	5.7	44	22	80	54	0.025	0.013	3.0	3.0	10
II	1	4.5	50	42	60	50	---	---	1.0	1.0	60
III	1	3.8	46	42	50	40	---	---	-	-	35
	2	12.0	48	20	140	125	---	---	-	-	50
IV 1/	1	24.0	-	-	140	125	0.740	0.570	-	-	-

1/ Two plies of type III, class 2 cloth.

3.4.1 Weave. The weave shall be plain, except that type III, class 2 and type IV, class 1 cloths shall be a herringbone twill.

3.4.2 Width. The width of the cloth shall be as specified (see 6.2) and shall be the minimum acceptable width inclusive of selvages.

3.5 Finish.

3.5.1 Type I. Type I cloth shall have a resin finish and the total nonfibrous materials, including resins, plasticizers and other substances, shall not exceed the amounts specified in table I.

3.5.2 Type II. Type II cloth shall have a non-chlorine retentive resin type finish and the material shall show no discoloration or loss of more than 10 percent of the original breaking strength of the cloth, when tested as specified in 4.4. The total nonfibrous materials including resins, plasticizers and other substances shall not exceed the amounts specified in table I.

3.5.3 Type III. Type III, class 1 cloth shall have a starch swing finish. Determination of the swing finish shall be made as specified in 4.4. Type III, class 2 cloth shall have a starch finish. The starch and protein content, including chloroform-soluble and water-soluble material for type III cloth, shall not exceed the amounts specified in table I when tested as specified in 4.4.

3.5.4 Type IV. Type IV cloth shall be made of two plies of type III, class 2 cloth. The two plies shall be joined by laminating the two plies together.

3.5.4.1 Laminating material. The laminating material shall be of a synthetic resin type, unaffected by wet drycleaning. There shall be no objectionable odor caused by the laminating materials.

3.5.4.1.1 Resistance to wet drycleaning. The laminated cloth shall show no evidence of delamination when subjected to three cycles of wet drycleaning when tested as specified in 4.4.

3.6 Acidity (pH). The pH value of the water extract of the finished cloth shall be not less than 5.5 nor more than 8.5 when tested as specified in 4.4.

3.7 Shrinkage (types I and II). The finished cloth for types I and II shall conform to the shrinkage requirements in table I when tested as specified in 4.4. The preshrinkage processes used shall not be identified by name or trademark, either on the cloth, ticket, or package.

3.8 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths each not less than 40 yards. Each length shall be put up in rolls as specified in PPP-P-1134.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.

3.9 Fiber identification. Each roll of cloth shall be labeled, ticketed or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

3.10 Workmanship. The finished cloth shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the point level specified.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.2 Examination of the end item. Examination of the end item shall be in accordance with 4.2.2.1 through 4.2.2.4.

4.2.2.1 Yard-by-yard examination. The entire yardage of each piece, but not more than 100 yards per piece, shall be examined on the face side. Defects as defined in Section I of Fed. Std. No. 4 which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.2.1.1 except that only those slubs and knots which exceed the maximum limits shown in figure 1 shall be scored. No contiguous linear yard or increment of one yard on the measuring device of the inspection machine, from any one roll within the sample shall be penalized more than 4 points.

The sample size shall be 20 rolls selected from 20 containers. The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds the following point values:

Type I, class 1 - 35.0 points
Type I, class 2 - 30.0 points
Types II, III, IV - 40.0 points

The lot shall be unacceptable if the points per 100 square yards of two or more individual rolls exceed the following values:

Type I, class 1 - 53.0 points
Type I, class 2 - 45.0 points
Types II, III, IV - 60.0 points

If no individual roll exceeds the point level, the lot shall be acceptable, with respect to this characteristic. If one roll exceeds the point level, a second sample of 20 rolls shall be examined only for individual roll quality examination. The lot

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.

shall be unacceptable if one or more rolls in the second sample exceeds the point level. For individual roll quality, examination of local defects only shall be scored. Overall type defects shall be scored only in determining compliance with the lot average quality requirements. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{total yards inspected}} = \text{Points per 100 square yards}$$

4.2.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension - one point.
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension - two points.
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension - three points.
For defects exceeding 9 inches in any dimension - four points.

The following defects, when present, shall be scored four points for each yard in which they occur:

Baggy, ridgy or wavy cloth.
Objectionable odor.
Width less than specified.
Poor dye penetration, mottled, streaky, or cloudy.

4.2.2.2 Examination for length.

4.2.2.2.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.2.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the ticket.

4.2.2.3 Examination for shade. During the yard-by-yard examination, each roll in the sample shall be examined for shade. Any roll in the sample off shade, shaded side to side, side to center, or end to end, shall be cause for rejection of the entire lot represented by the sample.

4.2.2.4 Examination for identification of preshrinkage process and for noncompliance with the Textile Fiber Products Identification Act. During the yard-by-yard examination, each roll in the sample shall be examined for these defects. The lot shall be unacceptable if two or more rolls in the sample contain identification of the preshrinkage process by name or trademark on the cloth or ticket, or are not labeled or ticketed in accordance with the Textile Fiber Products Identification Act.

4.3 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-1134 to determine that packaging, packing and marking requirements of section 5 of this specification are complied with.

4.4 Testing of the end item. The methods of testing specified in CCC-T-191, wherever applicable, and as listed in table II shall be followed. The physical and chemical values specified in section 3, except where otherwise specified, apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be as follows:

a. Four yards, full width of the finished cloth, for type II, class 1.
b. Three yards, full width of the finished cloth, for type I, classes 1 and 2.
c. One and one-half yards, full width of the finished cloth, for types III and IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.

The sample size shall be as follows:

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

The lot shall be unacceptable if one or more sample units fail to meet any requirement specified. All test reports shall contain the individual values utilized in expressing the final result.

TABLE II. Test method

Characteristic	Requirement paragraph	Test method	
Material (cotton yarn)	3.2.1	1200	2/
Weight	3.4	5041	
Yarns per inch	3.4	5050	
Breaking strength	3.4	5100	
Weave	3.4.1	Visual	1/
Swing finish (type III, class 1)	3.5.3	4.4.1	1/
Chlorine degradation (type II)	3.5.2	4.4.2	
Discoloration (type II)	3.5.2	4.4.3	
Stiffness	3.4	5202	
Colorfastness to:			
Laundering	3.3.2	5610	
Wet drycleaning	3.3.2	5622	
Crocking	3.3.2	5651	
Perspiration	3.3.2	5680	
Sulfur content	3.3	2020	
Nonfibrous materials:			
Type I	3.4 and 3.5.1	2/	
Type II	3.4 and 3.5.2	2/	
Type III	3.4 and 3.5.3	2611	2/
Shrinkage in laundering (types I and II)	3.4 and 3.7	5550	
Resistance to:			
Wet drycleaning for type IV (3 cycles)	3.5.4.1.1	5622	3/
Acidity (pH)	3.6	2811	

1/ One determination shall be made from each sample unit and the results reported as "pass" or "fail".
2/ The supplier shall submit a certificate of compliance for this characteristic. When required, the certificate shall be accompanied by actual test, inspection or other verifiable quality data.
3/ The specimen shall be examined for delamination. If the material exhibits delamination, it shall be cause for rejection.

4.4.1 Swing finish. A piece of type III, class 1 cloth, 10 by 10 inches shall be folded on a 45° bias without creasing. Hold the ends of the fold with both hands and pull on both ends of the cloth. Cloth having a swing finish will stretch or buckle.

4.4.2 Chlorine degradation (type II). A piece of cloth, not less than 20 inches long and full width, shall be laundered in accordance with method 5550 of CCC-T-191 and shall then be treated for one hour with 0.7 percent available chlorine solution with a temperature of 25 ± 1° C. The piece shall be rinsed in a sour solution of 0.014 percent acetic acid (0.05 ml. of 28 percent acetic acid per 100 ml. of water) for one minute at 80 ± 5° F (27 ± 3° C) and washed thoroughly until a control piece (undyed, desized, cotton cloth) shows no trace of chlorine. The piece shall be air-dried and then conditioned under Standard Conditions as defined in CCC-T-191. Eight pairs of specimens, four warpwise and four fillingwise, with each set having segments of the same

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.

yarns, shall be taken for tests. Four warp specimens and four filling specimens shall be tested in accordance with method 5100 of CCC-T-191. The remaining warp and filling specimens shall be oven dried at 105 to 110°C. for two hours and tested in the same manner as the first eight specimens. The difference in the average breaking strength of the respective warp and filling samples shall be computed and the percentage of loss determined.

4.4.3 Discoloration test (type II). A 4 by 4-inch piece of the chlorinated and conditioned sample of cloth from 4.4.2, with a piece of the unchlorinated cloth shall be oven dried at a temperature of 105 to 110°C. for two hours. Both pieces shall be removed from the oven and allowed to cool at room temperature. The chlorinated piece shall then be compared with the control cloth for discoloration. Any discoloration shall be cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. The cloth, put up as specified, shall be packaged in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Levels A, B, and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1134.

6. NOTES

6.1 Intended use. Types I, II and III cloth are used as interlining for clothing. Type II cloth is suitable for lining shirt collars, and type IV cloth (2 ply) is used when a high degree of stiffness is desired, as in the stand-up collar for cadet uniforms.

6.1.1 Type I, class 1. Used in:

Air Force	- Man's summer coat.
Navy	- Men's and women's uniforms.
	- Men's and women's overcoats (except pea coats).
Marine Corps	- Coats and belts.

6.1.2 Type I, class 2. Used in:

Air Force	- Wool serge coat.
Army	- Man's and woman's coats.
	- Man's overcoat.
Marine Corps	- Man's overcoat.
	- Man's coat.
	- Woman's coat.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Type and class of cloth required (see 1.2.1).
(c) Color of cloth required (see 3.3).
(d) Width of cloth required (see 3.4.2).
(e) When length is other than specified (see 3.8).
(f) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard sample, address the procuring office issuing the invitation for bids.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.

6.4 For civil agency procurement. When level B packaging is specified for civil agency procurement, the requirements of 5.1.1, level A packaging shall apply (see 5.1).

MILITARY CUSTODIANS:

Army - GL
Navy - SA
Air Force - 11

Review activities:

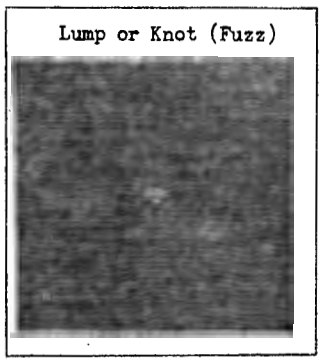
Army - MD
Navy - SA, MC
Air Force - 11, 82, 85

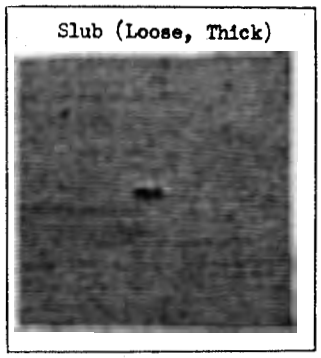
Preparing activity:

Army - GL

CIVIL AGENCIES INTEREST:

GSA-FSS





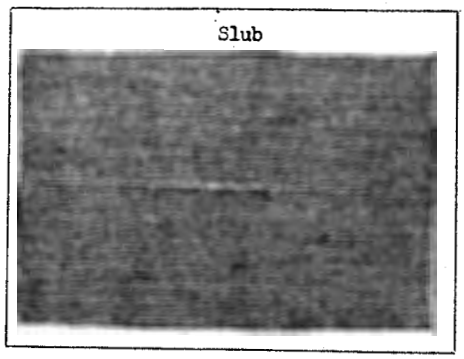


FIGURE 1. (Limits of Acceptability).

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein.
cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

CCC-C-438c Cloth, Buckram, Cotton

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
U. S. Army Natick Laboratories
Natick, Massachusetts 01760

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS

Commanding General
U. S. Army Natick Laboratories
ATTN: AMXRES-EQS
Natick, Massachusetts 01760

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.